

Problem

Let

$$x = \sqrt{a^2 + a + 1} - \sqrt{a^2 - a + 1}, \quad a \in \mathbb{R}.$$

Find all possible values of x .

Source: **Source:** Titu Andreescu

§1 Extracting a First Bound

Since

$$a^2 + |a| + 1 > |a|,$$

we have

$$\sqrt{a^2 + a + 1} + \sqrt{a^2 - a + 1} > 2|a|.$$

Rationalizing the expression for x gives

$$x = \frac{(a^2 + a + 1) - (a^2 - a + 1)}{\sqrt{a^2 + a + 1} + \sqrt{a^2 - a + 1}} = \frac{2a}{\sqrt{a^2 + a + 1} + \sqrt{a^2 - a + 1}}.$$

Hence

$$|x| < \frac{2|a|}{2|a|} = 1$$

for every $a \neq 0$, while $a = 0$ yields $x = 0$.

Therefore,

$$|x| < 1.$$

Thus every attainable value of x must belong to the interval

$$(-1, 1).$$

§2 Recovering a from x

Starting from

$$x = \sqrt{a^2 + a + 1} - \sqrt{a^2 - a + 1},$$

we isolate one radical:

$$x + \sqrt{a^2 - a + 1} = \sqrt{a^2 + a + 1}.$$

Squaring both sides yields

$$x^2 + 2x\sqrt{a^2 - a + 1} + a^2 - a + 1 = a^2 + a + 1,$$

and therefore

$$2x\sqrt{a^2 - a + 1} = 2a - x^2.$$

Squaring again, we obtain

$$4x^2(a^2 - a + 1) = (2a - x^2)^2.$$

Expanding and simplifying gives

$$4(x^2 - 1)a^2 = x^2(x^2 - 4),$$

hence

$$a^2 = \frac{x^2(x^2 - 4)}{4(x^2 - 1)}.$$

Since $a^2 \geq 0$, the right-hand side must be nonnegative:

$$\frac{x^2(x^2 - 4)}{x^2 - 1} \geq 0.$$

Because we already know that $|x| < 2$, we have

$$x^2 - 4 < 0.$$

Therefore the quotient is nonnegative only when

$$x^2 - 1 < 0,$$

that is,

$$|x| < 1.$$

§3 Showing Every Value Occurs

Let $x \in (-1, 1)$.

Then

$$x^2 - 1 < 0, \quad x^2 - 4 < 0,$$

so

$$\frac{x^2(x^2 - 4)}{4(x^2 - 1)} \geq 0.$$

Hence there exists a real number a satisfying

$$a^2 = \frac{x^2(x^2 - 4)}{4(x^2 - 1)}.$$

Reversing the preceding algebraic steps shows that this value of a indeed produces the prescribed value of x .

Thus every number in $(-1, 1)$ is attained.

§4 The Final Conclusion

Combining the necessity and sufficiency parts, the set of all possible values of x is

$$\boxed{(-1, 1)}.$$